

Question Paper

Exam Date & Time: 05-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026

SUBJECT: ICT 3123/IT_3123 SOFTWARE ENGINEERING

(SPL: INFORMATION TECHNOLOGY)

SOFTWARE ENGINEERING [ICT 3123]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

1A) Design a sequence diagram for an Online Banking System where a customer transfers money from their savings account to a friend's checking account. (5)

The system must:

- Validate login credentials and send an SMS code for second-step verification (one retry allowed before temporary account lock and security alert).
- Upon successful authentication, verify sufficient balance in the savings account.
- If funds are available, initiate the transfer by communicating concurrently with the Payment Processing Network (external) and the Transaction Logger (internal). The system should remain responsive during network delays.
- After payment confirmation, record the transaction, update both accounts, destroy temporary tokens, and send email/SMS notifications to both parties.

Explain your design choices regarding flow control, conditional paths, and communication patterns used.

1B) A banking software system has been designed with the following module interactions: Module A passes an account number (integer) to Module B for balance verification. Module C receives a customer data structure from Module D, but only uses the customer's credit score field. Module E sends a flag to Module F that determines whether to execute loan approval or rejection logic within Module F. Module G and Module H both access a global variable containing current interest rates. Module I directly modifies internal data variables of Module J and uses *goto* statements to jump into Module J's processing logic. Analyze each interaction by identifying the specific type of coupling present and rank them from most desirable to least desirable. Explain why the ranking you propose aligns with software engineering principles for maintainable system design. (3)

1C) In what scenarios would you recommend using a hybrid integration testing approach over a top-down or bottom-up approach? Explain your reasoning. (2)

2A) Analyze the merits and demerits of Client/Server Style, Repository Style, and Layered System. Synthesize a Hybrid Architecture Solution using these three styles for a distributed e-commerce system that must handle high traffic, ensure data consistency, and provide real-time inventory updates across multiple geographic regions. (5)

2B) A simple digital watch has a display and two buttons to set it, the A button and the B button. The watch has two modes of operation, display time and set time. In the display time mode, the watch displays hours and minutes, separated by a flashing colon. The set time mode has two submodes, set hours and set minutes. The A button selects modes. Each time it is pressed, the mode advances in the sequence: display, set hours, set minutes, display, etc. Within the submodes, the B button advances the hours or minutes once each time it is pressed. Buttons must be released before they can generate another event. Prepare a state diagram of the watch. (3)

- 2C) A software development team has created a module that handles user authentication, generates system logs, sends email notifications, and validates input data formats. When asked to justify their design, they argue that these functions are logically related because they all support the main application functionality. Analyze this design decision by identifying the type of cohesion currently present and propose a restructured design that would better align with software engineering best practices. (2)
- 3A) Design the test cases for the following code snippet using path testing. You are expected to follow the following steps to design an effective test case which have a high probability of revealing defects. (5)
1. Draw the CFG (Control Flow Graph)
 2. Find the Cyclomatic Complexity using three methods.
 3. Identify the independent paths (Basic Path Set)
 4. Derive test cases
- ```

int main()
{
 int percentage;
 char grade;
 do {
 printf("Enter an percentage :");
 scanf("%d", &percentage);

 if (percentage >=90)
 grade='A';
 else if (percentage >=80)
 grade='B';
 else if (percentage >=70)
 grade='C';
 else if (percentage >=60)
 grade='D';
 else if (percentage >=50)
 grade='E';
 else
 grade = 'U';

 printf("\n\nWant to check again (press Y/y for 'yes') :");
 scanf(" %c", &choice);
 } while (choice == 'Y' || choice == 'y');

 printf("\nYour grade is %c", grade);

 return 0;
}

```
- 3B) A software development project has identified the following risks: 30% probability that a key team member will leave (impact cost: \$15,000), 60% probability of requirement changes causing delays (impact cost: \$8,000), and 20% probability of integration issues with third-party systems (impact cost: \$25,000). The project manager must decide between reactive and proactive risk management approaches. Calculate the risk exposure for each identified risk using the Risk Exposure formula, categorize these risks according to the material's classification approaches, and develop an appropriate Risk Mitigation, Monitoring, and Management (RMMM) strategy for the highest priority risk. Justify whether a proactive or reactive approach would be more suitable for this project scenario. (3)
- 3C) Discuss the key phases of Requirements Engineering and analyze how each phase contributes to the success of a software project. (2)
- 4A) Based on the shipping cost calculator interface shown in Figure 4(A), design a comprehensive test plan for the *calculateShippingCost(weight, zone)* function using Equivalence Class Partitioning technique. Your test plan should include: (5)

- Identify all valid and invalid equivalence classes for both input parameters (weight and destination zone)
- Create test cases that cover each equivalence class
- Specify expected results for each test case

### Shipping Cost Calculator

#### Package Information

**Package Weight (kg):**

**Destination Zone:**

**CALCULATE**

#### Shipping Cost Results

**Total Shipping Cost:**  
**[\$[Calculated Amount]]**

#### Pricing Rules

| Weight Range   | Local (\$) | National (\$) | International |
|----------------|------------|---------------|---------------|
| 0.1 - 1.0 kg   | 5.00       | 10.00         | 25.00         |
| 1.1 - 5.0 kg   | 8.00       | 15.00         | 35.00         |
| 5.1 - 20.0 kg  | 12.00      | 25.00         | 50.00         |
| 20.1 - 50.0 kg | 20.00      | 40.00         | 80.00         |

#### Error Conditions

- Weight  $\leq 0.0$  kg or  $> 50.0$  kg: "Invalid weight"
- Invalid zone selection: "Invalid destination"
- Empty fields: "Please fill all required fields"

**calculateShippingCost(weight: double, zone: string) : double**

**Figure 4(A)**

Course Outcome: "CO4" / Cognitive Level: "Create"

- 4B) A university library provides services to students and faculty. Users can search for books, borrow books, and return books. A librarian manages the catalog and updates book records. The system also sends due-date reminders to users. Construct a use-case model for this Library Management System. Also, create an Activity Diagram for the process of Updating Book Records by the librarian. (3)

Course Outcome: "CO2" / Cognitive Level: "Create"

- 4C) Analyze the advantages and disadvantages of using Extreme Programming (XP) in a large organization compared to a small startup. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 5A) Create a class diagram for the following specification: Each instructor can teach several courses, (5)

and each course may be taught by more than one instructor. A student may register for many courses, and each course can have many students. A course is assigned to exactly one classroom, but a classroom may host different courses at different times. Each instructor is responsible for guiding multiple students, and a student may be guided by exactly one instructor. Draw the UML class diagram for this specification. Also, analyze whether the use of association class is possible with the relationship between Students and Courses. If yes, illustrate with a revised class diagram justifying such association class. If not, specify the reason.

Course Outcome: "CO2" / Cognitive Level: "Create"

- 5B) Analyze the general features and content of Software Configuration Management (SCM) and discuss its significance in maintaining project integrity and facilitating collaborative development. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5C) Compare verification and validation in the V-software process model, analyzing how each contributes to the quality of the software product. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

-----End-----

| Event Name | Subject |
|------------|---------|
|------------|---------|

058-MIT BLR-MAKEUP EXAM-2025-2026 IT\_3123:SOFTWARE ENGINEERING

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 4         |
| 2            | 2              | CO 2        | 8         |
| 3            | 4              | CO 4        | 5         |
| 4            | 5              | CO 5        | 3         |
| <b>Total</b> |                |             | <b>20</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT\_3123:SOFTWARE ENGINEERING

| #            | Level       | No of Questions | Marks (%)        |
|--------------|-------------|-----------------|------------------|
| 1            | Remember    | 0               | 0(0%)            |
| 2            | Understand  | 0               | 0(0%)            |
| 3            | Application | 0               | 0(0%)            |
| 4            | Analysis    | 3               | 7(35%)           |
| 5            | Create      | 0               | 0(0%)            |
| 6            | Evaluate    | 3               | 13(65%)          |
| <b>Total</b> |             | <b>6</b>        | <b>20 (100%)</b> |